Supplementary Table 3.1: Association between cancer mortality risk and selection for MHC I diversity Results of single-predictor phylogenetic logistic regressions explaining variance in **a** CMR or **b** ICM in function of selection for MHC I diversity, expressed as the ratio of non-synonymous to synonymous nucleotide substitution rates. Both models were weighted for the precision of the cancer mortality risk estimates (as indicated by the total number of individuals with RDI) selection for MHC I diversity estimate (based on the number of sequences used to estimate it).

	a. CMR			**b. ICM**		
	β (SE)	t-value	p-value	β (SE)	t-value	p-value
Intercept	-1.48 (1.04)	-1.42	0.1689	-0.67 (1.2)	-0.56	0.5793
Selection for MHC I diversity	-3.62 (2.13)	-1.70	0.1029	-5.89 (2.41)	-2.44	0.0232
Model Stats.	AICc = 62.85	n = 25	λ = 0.62	AICc = 65.99	n = 24	λ = 0.64

Supplementary Table 3.2: Association between cancer mortality risk and MHC I diversity at the 20 most positively selected sites (PSS) Results of single-predictor phylogenetic logistic regressions explaining variance in **a** CMR or **b** ICM in function of MHC I diversity at the 20 most positively selected sites (PSS), expressed as the ratio of non-synonymous to synonymous nucleotide substitution rates. Both models were weighted for the precision of the cancer mortality risk estimates (as indicated by the log total number of individuals with RDI).

	a. CMR			**b. ICM**		
	β (SE)	t-value	p-value	β (SE)	t-value	p-value
Intercept	-0.7_(0.7)	-1.00	0.3264	-1.07_(1.05)	-1.02	0.3190
Selection for MHC I diversity at PSS	-0.72_(0.22)	-3.26	0.0034	-0.65_(0.3)	-2.14	0.0436
Model Stats.	AICc = 63.3	n = 25	λ = 0.19	AICc = 71.63	n = 24	λ = 0.45

Supplementary Table 3.3: Association between cancer mortality risk and MHC class I gene family size Results of single-predictor phylogenetic logistic regressions explaining variance in **a** CMR or **b** ICM in function of MHC class I gene family size, expressed as the number of protein coding MHC class I genes. Both models were weighted for the precision of the cancer mortality risk estimates (as indicated by the log total number of individuals with RDI).

	a. CMR			**b. ICM**		
	β (SE)	t-value	p-value	β (SE)	t-value	p-value
Intercept	-1.73 (0.50)	-3.49	0.0007	-1.76 (0.55)	-3.19	0.0019
Log MHC class I gene family size	-0.51 (0.14)	-3.68	0.0004	-0.47 (0.17)	-2.76	0.0068
Model Stats.	*AICc = 234.99*	*n = 106*	*λ = 0.55*	*AICc = 261.27*	*n = 100*	*λ = 0.33*

Supplementary Table 3.4: Association between cancer mortality risk and selection for MHC I diversity with body mass and life expectancy as potential confounding effects Results of phylogenetic logistic regressions explaining variance in **a** CMR or **b** ICM in function of body mass, life expectancy and selection for MHC I diversity at all sites (top model) and at the 20 most positively selected sites (PSS), expressed as the ratio of non-synonymous to synonymous nucleotide substitution rates. Both models were weighted for the precision of the cancer mortality risk estimates (as indicated by the log total number of individuals with RDI) .

	a. CMR			**b. ICM**		
	β (SE)	t-value	p-value	β (SE)	t-value	p-value
Intercept	1.22 (4.39)	0.28	0.7839	5.84 (4.84)	1.21	0.2414
Log body mass	0.05 (0.09)	0.52	0.6117	0.14 (0.11)	1.31	0.2036
Log life expectancy	-0.35 (0.54)	-0.66	0.5191	-0.82 (0.59)	-1.40	0.1781
Selection for MHC I diversity	-3.35 (2.33)	-1.44	0.1645	-5.62 (2.51)	-2.24	0.0370
Model Stats.	AICc = 71.73	n = 25	λ = 0.7	AICc = 69.05	n = 24	λ = 0.74
Intercept	3.88 (3.66)	1.06	0.3014	8.54 (4.72)	1.81	0.0857
Log body mass	0.05 (0.07)	0.65	0.5216	0.13 (0.10)	1.41	0.1750
Log life expectancy	-0.57 (0.42)	-1.34	0.1955	-1.15 (0.54)	-2.12	0.0465
Selection for MHC I diversity at PSS	-0.69 (0.24)	-2.89	0.0088	-0.69 (0.30)	-2.32	0.0307
Model Stats.	AICc = 71.87	n = 25	λ = 0.43	AICc = 73.01	n=24	λ = 0.60

Supplementary Table 3.5: Association between cancer mortality risk and MHC class I gene family size with body mass and life expectancy as potential confounding effects Results of phylogenetic logistic regressions explaining variance in a CMR or b ICM in function of body mass, life expectancy and MHC class I gene family size, expressed as the number of protein coding MHC class I genes. Both models were weighted for the precision of the cancer mortality risk estimates (as indicated by the log total number of individuals with RDI).

	a. CMR			**b. ICM**		
	β (SE)	t-value	p-value	β (SE)	t-value	p-value
Intercept	-2.59 (1.50)	-1.73	0.0870	-0.93 (1.86)	-0.50	0.6182
Log life expectancy	0.10 (0.18)	0.58	0.5633	-0.11 (0.21)	-0.52	0.6008
Log body mass	0.00 (0.04)	-0.06	0.9523	0.03 (0.05)	0.62	0.5392
Log MHC class I gene family size	-0.50 (0.14)	-3.58	0.0005	-0.47 (0.18)	-2.68	0.0086
Model Stats.	*AICc = 245.61*	*n = 106*	*λ = 0.56*	*AICc = 271.09*	*n = 100*	*λ = 0.37*

Supplementary Table 3.6: Association between cancer mortality risk and selection for MHC I diversity, controlling for diet Results of phylogenetic logistic regressions explaining variance in **a** CMR or **b** ICM in function of mammal-based diet (i.e. mammals constituting the primary/secondary food item) and MHC I diversity, expressed as the ratio of non-synonymous to synonymous nucleotide substitution rates. Both models were weighted for the precision of the cancer mortality risk estimates (as indicated by the log total number of individuals with RDI).

	a. CMR			**b. ICM**		
	β (SE)	t-value	p-value	β (SE)	t-value	p-value
(Intercept)	-1.81 (1.19)	-1.53	0.1413	-0.55 (1.41)	-0.39	0.6991
Mammalian prey	0.38 (0.61)	0.63	0.5374	-0.17 (0.75)	-0.23	0.8204
Selection for MHC I diversity	-3.03 (2.37)	-1.28	0.2152	-6.16 (2.66)	-2.32	**0.0307**
Model Stats.	AICc = 64.77	n = 25	λ = 0.6	AICc = 67.99	n = 24	λ = 0.7
	β (SE)	t-value	p-value	β (SE)	t-value	p-value
(Intercept)	-0.85 (1.01)	-0.84	0.4101	-0.99 (1.37)	-0.72	0.4800
Mammalian prey	0.04 (0.55)	0.08	0.9387	-0.13 (0.75)	-0.17	0.8688
Selection for MHC I diversity at PSS	-0.68 (0.29)	-2.30	0.0311	-0.67 (0.37)	-1.78	0.0891
Model Stats.	AICc = 65.89	n = 25	λ = 0.31	AICc = 73.63	n = 24	λ = 0.53

Supplementary Table 3.7: Association between cancer mortality risk and MHC class I gene family size, controlling for diet Results of phylogenetic logistic regressions explaining variance in **a** CMR or **b** ICM in function of mammal-based diet (i.e. mammals constituting the primary/secondary food item) and MHC class I gene family size, expressed as the number of protein coding MHC class I genes. Both models were weighted for the precision of the cancer mortality risk estimates (as indicated by the log total number of individuals with RDI).

	a. CMR			**b. ICM**		
	β (SE)	t-value	p-value	β (SE)	t-value	p-value
(Intercept)	-1.83 (0.49)	-3.75	0.0003	-1.74 (0.53)	-3.30	0.0014
Mammalian prey	0.36 (0.23)	1.55	0.1250	0.32 (0.27)	1.19	0.2385
Log MHC class I gene family size	-0.48 (0.14)	-3.47	0.0008	-0.48 (0.17)	-2.82	0.0058
Model Stats.	AICc = 236.08	n = 106	λ = 0.48	AICc = 263.14	n = 100	λ = 0.22

Supplementary Table 3.8: Association between MHC I and life-history traits Results of phylogenetic regressions exploring the association between selection for **a** MHC I diversity, **b** MHC I diversity at the 20 most positively selected sites (PSS) or **c** MHC I gene family size and body mass, adult life expectancy and 80% adult longevity. Selection for MHC I diversity was expressed as the ratio of non-synonymous to synonymous nucleotide substitution rates. Models with life expectancy and longevity 80% were weighted for the log number of individuals used to estimate the longevity metrics.

	a. Selection for MHC I diversity			**b. Selection for MHC I diversity at PSS**			**c. MHC I gene family size**		
	β (SE)	t-value	p-value	β (SE)	t-value	p-value	β (SE)	t-value	p-value
Intercept	0.38 (0.08)	4.67	0.0001	2.87 (0.65)	4.38	0.0002	2.49 (0.52)	4.81	<0.0001
Log body mass	0.01 (0.01)	1.47	0.1537	0.04 (0.06)	0.76	0.4548	-0.02 (0.03)	-0.62	0.5335
Model Stats.	AICc = -48.67	n = 28	λ = 0.77	AICc = 59.88	n = 28	λ = 0.78	AICc = 178.13	n = 125	λ = 0.90
	β (SE)	t-value	p-value	β (SE)	t-value	p-value	β (SE)	t-value	p-value
Intercept	0.22 (0.11)	2.06	0.0491	3.02 (0.97)	3.13	0.0043	2.6 (0.43)	6.09	<0.0001
Log life expectancy	0.06 (0.04)	1.80	0.0835	-0.05 (0.31)	-0.17	0.8698	-0.15 (0.1)	-1.49	0.1384
Model Stats.	AICc = -52.95	n = 28	λ = 0.65	AICc = 54.5	n = 28	λ = 0.78	AICc = 172.67	n = 125	λ = 0.91
	β (SE)	t-value	p-value	β (SE)	t-value	p-value	β (SE)	t-value	p-value
Intercept	-0.43 (0.27)	-1.57	0.1288	1.42 (2.54)	0.56	0.5801	3.19 (1.00)	3.19	0.0018
Log 80% adult longevity	0.09 (0.03)	3.07	0.0051	0.17 (0.29)	0.61	0.5502	-0.09 (0.11)	-0.76	0.4492
Model Stats.	AICc = -54.37	n = 27	λ = 0.69	AICc = 56.11	n = 27	λ = 0.78	AICc = 170.27	n = 122	λ = 0.89